

TAMAR NOSELIDZE

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SUMMARY

Quantum Information Master's Student at Sorbonne University, good knowledge of quantum dynamics combined with advanced machine learning and high-performance computing. Experienced in using convex optimization, scientific computing, and PyTorch pipelines to model complex quantum systems. Eager to apply my background in AI and mathematical optimization to autonomous quantum error correction and fault-tolerant architectures. Looking for a 6-month M2 internship to contribute to quantum computing solutions.

RESEARCH EXPERIENCE

Quantum Information Researcher

2026 – Present, Campinas (*remote*) & Paris

UNICAMP & Sorbonne University

- **Quantum Nonlocality & SVMs:** Engineered a novel SVM pipeline to classify Local Hidden States (LHS) and quantum steerability by compressing 2-qubit density matrices into a 9-dimensional Fano feature representation. Extracted a novel algebraic formula to capture quantum nonlocality, overcame traditional LP bottlenecks by implementing convex optimization pipelines using MOSEK; and presented finalized findings to the UNICAMP physics department.
<https://github.com/TamarNoselidze/SVM-Nonlocality-Boundary>
- **Multipartite Bell Scenarios** (*paper in preparation*): Defining high-dimensional polytopes and applying vertex annotation (using PORTA, PANDA) to discover novel Bell inequalities. Using convex optimization (in CVXPY), linear programming, and see-saw techniques to model complex quantum correlations.
- **Quantum Catalysis** (*paper in preparation*): Investigating the catalytic power of quantum states across different dimensions, analyzing entanglement as a novel thermodynamic and informational resource.

AI Researcher

2024 – 2025, Prague, Czech Republic

<https://dspace.cuni.cz/handle/20.500.11956/200886>

Charles University

- **Adversarial AI:** Achieved >75% attack success rate on Vision Transformer and CNN models by engineering adversarial patch attacks using PyTorch.
- **HPC Simulation:** Designed and implemented reproducible Python-based machine learning pipelines, consuming over 500 GPU days on the MetaCentrum cluster.
- **Scientific Workflow:** Collaborated in an agile research environment, using Git for version control and documenting experimental results for academic publication.
- **Model Optimization:** Integrated pre-trained models and applied optimization techniques to improve reliability and performance on large image datasets.

EDUCATION

Master's of Quantum Information (QI)

Sorbonne University • Paris, France • 2025–Present

- Coursework: Quantum Algorithms, Quantum Dynamics, Quantum Information Theory, Mathematical Algorithms & Complexity, Data Science & Statistical Learning, Photonic Quantum Computing, Quantum Cryptography.

Bachelor of Computer Science – Artificial Intelligence

Charles University • Prague, Czech Republic • 2021–2025

- Focused on AI and Machine Learning, with rigorous mathematical background.
- Coursework: Linear Algebra, Probability & Statistics, Graph Theory, Mathematical Analysis, Machine Learning, Computer Vision, NLP.

SKILLS

- **Quantum & Optimization:** Convex Optimization, Linear Programming, See-Saw Technique, Hybrid ML–Quantum Algorithms, Bell Nonlocality.
- **Optimization & Scientific Tooling:** MOSEK, CVXPY, PORTA, PANDA, RANDA, NumPy, Pandas.
- **Development & HPC:** Git/GitHub, Docker/Singularity (Containerization), Bash, Linux, HPC Workflows (MetaCentrum).
- **Programming Languages:** Python (Advanced), Julia, Java, C#, Haskell, MATLAB.
- **Machine Learning:** PyTorch, Hugging Face Transformers, SVMs, Scikit-learn, Computer Vision, Adversarial Robustness.
- **Spoken Languages:** English (Fluent), Georgian (Native), French (Beginner).